Yatharth Thapa

Location - Dehradun, Uttarakhand, India
Integrated B.Tech & M.Tech - Information Technology
Indian Institute Of Information Technology, Gwalior

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LinkedIn | Portfolio | Github
LeetCode | Codeforces

Summary

Full-Stack Engineer with **1.5+** years of experience building both scalable backend services **(Node.js, NestJS, PostgreSQL, Redis, AWS)** and high-performance frontends using **React, Next.js, and TypeScript**. I've worked across **microservices**, **distributed systems**, and **modern UI architectures**, focusing on clean API design, performance, **Core Web Vitals**, and smooth **state management**. I enjoy building fast, reliable, and user-friendly products end-to-end, with attention to **scalability**, and **production-grade** standards.

Education

- **Indian Institute Of Information Technology (IIIT), Gwalior, India**
  B.Tech + M.Tech in Information Technology | CGPA: 7.1 — July 2019 – July 2024

Experience

- **Flywl** — *February 2025 – October 2025*
  Full Stack Engineer — Bangalore - India
  - Developed scalable **REST APIs** using **NestJS** + **TypeORM** (improved backend throughput by **35%**)
  - Optimized **PostgreSQL** using **indexing**, **query-plan tuning** (supported **99.9%** uptime and faster operations)
  - Added **Redis caching** for frequent news queries (reduced DB load by **50%** and improved API latency by **30%**)
  - Integrated **S3**, **CloudWatch**, and **SNS** for storage, monitoring, and asynchronous notifications
  - Implemented **code-splitting**, **RSC(React Server Components**, and bundle optimizations (reduced client-side js size by **35%**)
  - Used Git for version control and applied strong problem-solving and data structures fundamentals in development tasks.
  - Rebuilt UI components using **memoization**, **virtualization**, and **Profiler insights**; (improved render performance)
  - Set up a predictable data layer with **Redux Toolkit** + **React Query** (reduced unnecessary re-renders by **40%**)
  - Improved page speed using **dynamic imports**, **lazy loading**, and **code cleanup** (reduced bundle sizes across pages)
- **SL Technocrats** — *May 2024 – January 2025*
  Full Stack Engineer — Hyderabad - India
  - Built a **Dockerized PDF-to-DOC** microservice (improved medical document processing speed and reliability)
  - Added **rate limiting**, **schema validation**, and **JWT auth** (increased API reliability and security by **30%**)
  - Integrated Orthanc PACS with React-based OHIF Viewer (optimized DICOM fetching by **60%** faster rendering)
  - Implemented **client-side caching** + **invalidation strategies** (reduced redundant API calls by **50%**)
  - Built reliable **client–server** communication using **REST** and **WebSockets** (enabled real-time UI updates)
  - Applied **GDCM-based** DICOM compression (reduced file size and upload latency by 20%)

Projects

- **AI Learning Assistant — An AI powered learning and practicing platform** — ***LIVE / DEMO***
  Google gemini, React.js, Tailwindcss, ContextAPI, JWT, Mongoose, Bcrypt, Express, Mongodb — **Github**
  - Dummy credentials — email : **dummy@mail.com** password: **password**
  - Developed a full-stack AI learning assistant using React.js, Node.js, Express.js, and MongoDB, implementing complete CRUD workflows, responsive UI, and scalable RESTful API integration.
  - Integrated Google Gemini AI services to enable PDF-based summarisation, AI chat in document context, flashcard and quiz generation, handling asynchronous data flow and efficient frontend–backend communication.
  - Built reusable React components with Redux state management, implemented secure API handling, and worked with NoSQL databases and cloud services, ensuring clean code, version control using Git.
- **GitHub-ShortURL — Serverless URL Shortener Using Git Internals** — ***LIVE / DEMO***
  React.js, Git internals, Github REST API's — **Github**
- **Remote Code Executor — Collaborate in real time on a single, shared coding interface** — ***LIVE / DEMO***
  Socket.io, Express.js, Morgan, Docker — **Github**

Technical Skills

- **Programming**: C/C++, Python, JavaScript, TypeScript, SQL, Bash
- **Frontend**: React, Next.js, Svelte, TailwindCSS, Redux Toolkit, Redux Thunk, React Query, Zustand, Axios, Component Architecture, Responsive Design, Performance Optimization
- **Backend**: Node.js, Express.js, NestJS, FastAPI, REST APIs, Microservices, TypeORM, Sequelize, Knex.js, PostgreSQL, MongoDB, Message Queues (Kafka, RabbitMQ), Authentication (JWT, OAuth2.0, RBAC), Caching, API Security, Rate Limiting, Load Balancing
- **System & DevOps** : Docker, Git, GitHub Actions, CI/CD, Nginx, SSL, Monitoring (Grafana, Prometheus, CloudWatch), API Design & Best Practices, CAP Theorem, Database Sharding & Replication
- **Core CS**: Data Structures & Algorithms, System Design, Distributed Systems, Microservices, Concurrency, TCP/UDP, Big-O Analysis, Query Optimization, Event-Driven Architecture

Awards and Achievements

- **GATE Qualified, JEE Mains Rank**: 15,453
- **Knight Badge Leetcode**: Top 10% in contest rankings, solved over 750+ DSA questions
- **JPMorgan Chase & Co. - Software Engineering Job Simulation**: Kafka Integration, H2 Integration, REST API Integration, REST API Controller
- **Wallmart Global Tech - Advanced Software Engineering Job Simulation**: Advanced Data Structures, Software Architecture, Relational Database Design, Data Munging